MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics **WEEKLY CHAPTER TEST – 01 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 12: Surface Areas and Volumes** **MM: 20**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions & Assertion-Reason MATHS MANTHAN | ($4 \times 1 = 4$ Marks)

Q1. The diameter of a metallic sphere is 6 cm. It is melted and drawn into a wire of diameter 2 mm. The length of the wire is: [1]

(A) 12 m (B) 18 m (C) 36 m (D) 66 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 36 m** [0.5 Mark]

Given: Radius of sphere $R = \frac{6}{2} = 3$ cm. Wire radius $r = \frac{2}{2} = 1$ mm $= 0.1$ cm $= \frac{1}{10}$ cm.
We know that volume remains conserved during melting and recasting:

Formula / Theorem Used: Volume of Sphere $= \frac{4}{3}\pi R^3$ and Volume of Cylindrical Wire $= \pi r^2 h$

Equating the two volumes (h = length of wire in cm): [0.5 Mark]

$$\frac{4}{3}\pi(3)^3 = \pi\left(\frac{1}{10}\right)^2 h \implies \frac{4}{3} \times 27 = \frac{1}{100}h \implies 36 = \frac{h}{100}$$

$$h = 36 \times 100 = 3600 \text{ cm} = \frac{3600}{100} \text{ m} = \boxed{\textbf{36 m}}$$

Q2. A metallic solid sphere of radius 10.5 cm is melted and recast into small cones, each of radius 3.5 cm and height 3 cm. The number of such cones formed is: [1]

(A) 63 (B) 126 (C) 21 (D) 130

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 126** [0.5 Mark]

Given: Sphere radius $R = 10.5$ cm $= \frac{21}{2}$ cm; Cone radius $r = 3.5$ cm $= \frac{7}{2}$ cm, height $h = 3$ cm.
We know the volume formulas and conservation principle (n = number of cones):

Formula / Theorem Used: $n = \dfrac{\text{Volume of Sphere}}{\text{Volume of One Cone}} = \dfrac{\frac{4}{3}\pi R^3}{\frac{1}{3}\pi r^2 h}$

Cancelling $\frac{1}{3}\pi$ from numerator and denominator and substituting values: [0.5 Mark]

$$n = \frac{4R^3}{r^2 h} = \frac{4 \times 10.5 \times 10.5 \times 10.5}{3.5 \times 3.5 \times 3} = \frac{4 \times 3 \times 3 \times 10.5}{3} = 4 \times 3 \times 10.5 = \boxed{\textbf{126}}$$

Q3. A solid is hemispherical at the bottom and conical above. If the surface areas of the two parts are equal, then the ratio of its radius and the height of its conical part is: [1]

(A) $1:\sqrt{3}$ (B) $1:1$ (C) $\sqrt{3}:1$ (D) $1:2$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: $\boxed{\textbf{(A) } 1:\sqrt{3}}$ [0.5 Mark]

Given: Common base radius $= r$, slant height of cone $= l$, vertical height of cone $= h$, and CSA of hemisphere = CSA of cone.

We know the curved surface area and slant-height formulas:

Formula / Theorem Used: CSA of Hemisphere $= 2\pi r^2$, CSA of Cone $= \pi rl$, $l^2 = r^2 + h^2$

Equating the curved surface areas: [0.5 Mark]

$$2\pi r^2 = \pi rl \implies 2r = l \quad (\text{dividing by } \pi r > 0)$$

Substituting $l = 2r$ into $r^2 + h^2 = l^2$:

$$r^2 + h^2 = (2r)^2 = 4r^2 \implies h^2 = 3r^2 \implies h = \sqrt{3}\,r \implies \frac{r}{h} = \frac{1}{\sqrt{3}} \implies \boxed{\mathbf{1:\sqrt{3}}}$$

Q4. Assertion (A): Three cubes each of volume 8 cm^3 are joined end-to-end to form a cuboid. The total surface area of the resulting cuboid is 56 cm^2.

Reason (R): If n cubes each of volume a^3 are joined end-to-end, the total surface area of the cuboid is $2(2n+1)a^2$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: $\boxed{\textbf{(A)}}$ [0.5 Mark]

Verification of (R) & (A): When n cubes of side a are joined end-to-end, the resulting cuboid has $L = na$, $B = a$, $H = a$. [0.5 Mark]

$$\text{TSA} = 2(LB + BH + HL) = 2(na^2 + a^2 + na^2) = 2(2n+1)a^2 \quad (\text{Reason (R) is true.})$$

For $n = 3$ and $a^3 = 8 \implies a = 2$ cm: TSA $= 2(2(3)+1)(2)^2 = 2 \times 7 \times 4 = 56$ cm^2. Both (A) and (R) are true, and (R) correctly explains (A).

SECTION B: Short Answer Questions

MATHS MANTHAN | **(2 × 2 = 4 Marks)**

Q5. A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Step 1: Dimensions & Formula Declaration [1 Mark]

Given: Diameter $= 14$ cm $\implies$ common radius $r = \frac{14}{2} = 7$ cm.

Total height of vessel $= 13$ cm $\implies$ height of cylinder $h = 13 - r = 13 - 7 = 6$ cm.

We know the inner curved surface area formulas:

Formula / Theorem Used: Inner SA = CSA of Cylinder + CSA of Hemisphere $= 2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

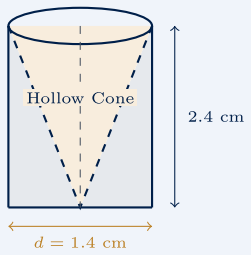
Step 2: Substitution & Arithmetic [1 Mark]

Substituting $\pi = \frac{22}{7}$, $r = 7$ cm, and $h = 6$ cm:

$$\text{Inner SA} = 2 \times \frac{22}{7} \times 7 \times (6+7) = 44 \times 13 = \boxed{\mathbf{572}\ \text{cm}^2}$$

Q6. From a solid cylinder of height 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Step 1: Dimensions & Slant Height [1 Mark]

Given: Height $h = 2.4$ cm, diameter $d = 1.4$ cm $\implies r = 0.7$ cm.

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{(0.7)^2 + (2.4)^2} = \sqrt{0.49 + 5.76} = \sqrt{6.25} = 2.5$ cm.

We know that the exposed surfaces are outer cylinder CSA, top/bottom solid circle base, and inner cone CSA:

Formula / Theorem

Used: $\text{TSA} = 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l)$

Step 2: Substitution & Simplification [1 Mark]

$$\text{TSA} = \frac{22}{7} \times 0.7 \times (2 \times 2.4 + 0.7 + 2.5)$$

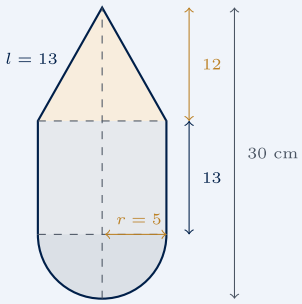
$$= 2.2 \times (4.8 + 3.2) = 2.2 \times 8 = \boxed{\mathbf{17.6}\ \text{cm}^2}$$

SECTION C: Short Answer Questions

MATHS MANTHAN | (2 × 3 = 6 Marks)

Q7. A toy is in the shape of a right circular cylinder with a hemisphere on one end and a cone on the other. The radius and height of the cylindrical part are 5 cm and 13 cm respectively. The radii of the hemispherical and conical parts are the same as that of the cylindrical part. Find the surface area of the toy if the total height of the toy is 30 cm. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Decomposing Dimensions [1 Mark]

Common radius $r = 5$ cm, cylinder height $h_{\text{cyl}} = 13$ cm, hemisphere height $= r = 5$ cm.

Height of cone $h_{\text{cone}} = 30 - (13 + 5) = 12$ cm.

Slant height of cone: $l = \sqrt{r^2 + h_{\text{cone}}^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ cm.

Step 2: Combining Surface Areas [1 Mark]

Formula / Theorem

Used: $\text{TSA} = \text{CSA}_{\text{hemi}} + \text{CSA}_{\text{cyl}} + \text{CSA}_{\text{cone}} = 2\pi r^2 + 2\pi rh_{\text{cyl}} + \pi rl = \pi r(2r + 2h_{\text{cyl}} + l)$

Step 3: Substitution & Calculation [1 Mark]

$$\text{TSA} = \frac{22}{7} \times 5 \times (10 + 26 + 13)$$

$$= \frac{22}{7} \times 5 \times 49 = 22 \times 5 \times 7 = \boxed{\mathbf{770}\ \text{cm}^2}$$

Q8. A cylindrical vessel with internal diameter 10 cm and height 10.5 cm is full of water. A solid cone of base diameter 7 cm and height 6 cm is completely immersed in water. Find the volume of water: (i) displaced out of the cylinder, (ii) left inside the cylindrical vessel. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Cylinder: $R = \frac{10}{2} = 5$ cm, $H = 10.5$ cm. Cone: $r = \frac{7}{2} = 3.5$ cm, $h = 6$ cm.

Formula / Theorem Used: Volume of Cylinder $= \pi R^2 H$ and Volume of Cone $= \frac{1}{3}\pi r^2 h$

Step 1: (i) Water Displaced Out (= Immersed Cone Volume) [1.5 Marks]

$V_{\text{displaced}} = \frac{1}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 6 = 11 \times 7 = \boxed{\mathbf{77\ cm^3}}$.

Step 2: (ii) Volume of Water Left Inside Vessel [1.5 Marks]

$V_{\text{cylinder}} = \frac{22}{7} \times 5 \times 5 \times \frac{21}{2} = 11 \times 25 \times 3 = 825\ \text{cm}^3$.

$V_{\text{left}} = V_{\text{cylinder}} - V_{\text{displaced}} = 825 - 77 = \boxed{\mathbf{748\ cm^3}}$.

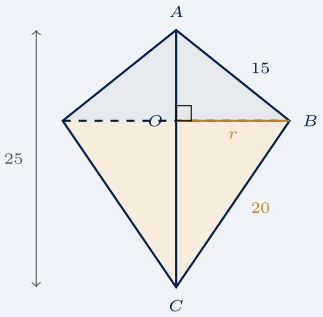
SECTION D: Long Answer / HOTS Question — MATHS MANTHAN | (6 Marks)

Q9. A right-angled triangle whose perpendicular sides are 15 cm and 20 cm is made to revolve about its hypotenuse.
(i) Determine the common base radius of the double cone so formed.
(ii) Calculate the total volume of the double cone. (Use $\pi = 3.14$)
(iii) Find the total surface area of the double cone exposed to the exterior. **[6]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 6 Marks]

Step 1: Hypotenuse & (i) Common Radius r [2 Marks]

In right $\triangle ABC$ ($\angle B = 90°$), $AB = 15$ cm and $BC = 20$ cm.
By Pythagoras Theorem: $AC = \sqrt{15^2 + 20^2} = \sqrt{225 + 400} = \sqrt{625} = 25$ cm.
Revolving $\triangle ABC$ about hypotenuse AC creates two cones sharing radius $r = BO \perp AC$. Using area of $\triangle ABC$:



$$\frac{1}{2} \times AB \times BC = \frac{1}{2} \times AC \times BO$$

$$15 \times 20 = 25 \times r \implies r = \frac{300}{25} = \boxed{\mathbf{12\ cm}}$$

Step 2: (ii) Total Volume of Double Cone [2 Marks]

Formula / Theorem Used: $V = \frac{1}{3}\pi r^2 h_1 + \frac{1}{3}\pi r^2 h_2 = \frac{1}{3}\pi r^2 (h_1 + h_2) = \frac{1}{3}\pi r^2 \cdot AC$

Substituting $\pi = 3.14$, $r = 12$ cm, and $h_1 + h_2 = AO + OC = AC = 25$ cm:

$$V = \frac{1}{3} \times 3.14 \times 12^2 \times 25 = 3.14 \times 48 \times 25 = 3.14 \times 1200 = \boxed{\mathbf{3768\ cm^3}}$$

Step 3: (iii) Exterior Total Surface Area [2 Marks]

Slant heights are $l_1 = AB = 15$ cm and $l_2 = BC = 20$ cm.

Formula / Theorem Used: TSA $= \pi r l_1 + \pi r l_2 = \pi r (l_1 + l_2)$

$$\text{TSA} = 3.14 \times 12 \times (15 + 20) = 37.68 \times 35 = \boxed{\mathbf{1318.8\ cm^2}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★